

GENERAL CATALOG 2024

Edition V2 — Rock Wool Sandwich Panel Systems

ENVELOPE ENGINEERING

Fire Safety, Thermal Efficiency, and High-Performance Acoustic Solutions

Manufacturer:

Vertex Rock Systems (Premium Constructive Systems)

Material Specification:

Structural insulating core made of high-density basaltic rock wool ($\rho \geq 100 \text{ kg/m}^3$), with fibers oriented perpendicularly to the galvanized and pre-painted steel facings, bonded using a high mechanical strength two-component polyurethane adhesive.

1 Introduction and Core Technology

Modern architectural and industrial design demands materials that not only meet aesthetic and structural requirements but also guarantee maximum active and passive protection. The basaltic rock wool core of our panels represents the pinnacle of mineral insulation technology, significantly outperforming synthetic or plastic cores such as expanded polystyrene (EPS), polyurethane (PUR), or polyisocyanurate (PIR).

1.1 Fire Performance and Non-Combustibility

Unlike organic polymers (PUR/PIR/EPS) that undergo thermolysis, melting, and toxic gas emission at relatively low temperatures, basaltic rock wool is intrinsically non-combustible. Its melting point exceeds 1000°C.

- **Reaction to Fire Classification (Euroclasses):** Our standard panels are classified as **A2-s1, d0** according to the *EN 13501-1* standard. This signifies that they do not contribute to fire growth, smoke emission is virtually non-existent, and there is no release of flaming droplets or particles. The pure mineral core itself achieves the maximum non-combustibility rating of **A1**.

1.2 Outstanding Acoustic Performance

The multi-directional, open, and porous structure of rock wool acts as an exceptionally efficient sound wave dampener. While closed-cell plastic cores (such as PIR) exhibit poor acoustic performance due to their rigidity and low mass, our rock wool panels combine mass (densities ranging from 100 to 120 kg/m³) and porosity to:

- Maximize the overall weighted sound reduction index (R_w).
- Dissipate sound energy through viscous friction within the mineral fiber matrix.

1.3 Sustainability and Long-Term Durability

- **Dimensional Stability:** Rock wool does not experience significant contraction or expansion due to extreme thermal fluctuations. This ensures that panel joints remain airtight throughout the building's lifespan, eliminating thermal bridging.
- **Chemical Inertness:** It is chemically neutral, does not promote corrosion of the steel facings, and is immune to attacks by microorganisms, mold, and rodents. Furthermore, it does not absorb water via capillary action due to hydrophobic oil treatments applied during the spinning process.
- **Life Cycle:** It is a 100% recyclable material manufactured from basaltic rock abundant in the Earth's crust, actively contributing to LEED, BREEAM, and DGNB green building certifications.

2 Product Range

2.1 Wall Panel: “RockFacade”

Designed for architectural and industrial external cladding, offering options for both visible and concealed fastening systems. Its engineered design ensures continuous insulation and outstanding resistance to wind loads.

- **Recommended Uses:** Cladding for industrial warehouses, logistics centers, commercial buildings, data centers, and high-security fire compartmentation walls.
- **Standard Useful Widths:** 1000 mm and 1150 mm.
- **Surface Profiles:** Flat, micro-ribbed, and standard box profile.
- **Steel Substrates:** Outer and inner skin made of galvanized steel according to *EN 10346* with standard thicknesses of 0.5 mm, 0.6 mm, and 0.7 mm. Coatings include standard 25 μm Polyester, PVDF, or HDX for highly corrosive environments.

2.1.1 Thicknesses and Typical Thermal Transmittance (*U*-Value)

- Thickness: 50 mm $\rightarrow U = 0.78 \text{ W}/(\text{m}^2 \cdot \text{K})$
- Thickness: 80 mm $\rightarrow U = 0.50 \text{ W}/(\text{m}^2 \cdot \text{K})$
- Thickness: 100 mm $\rightarrow U = 0.40 \text{ W}/(\text{m}^2 \cdot \text{K})$
- Thickness: 120 mm $\rightarrow U = 0.34 \text{ W}/(\text{m}^2 \cdot \text{K})$
- Thickness: 150 mm $\rightarrow U = 0.27 \text{ W}/(\text{m}^2 \cdot \text{K})$
- Thickness: 200 mm $\rightarrow U = 0.20 \text{ W}/(\text{m}^2 \cdot \text{K})$

2.2 Roof Panel: “RockRoof”

A high-performance sandwich panel system designed for pitched roofs, featuring a profiled external geometry that provides optimal structural rigidity against snow, maintenance, and wind loads.

- **Profile Design and Overlap:** Engineered with a 2, 3, or 5-rib configuration. It includes an steel cap-profile (joint-cover) that mechanically seals the longitudinal panel joint, shielding the fasteners from rain and preventing water ingress.
- **Minimum Installation Pitch:**
 - $\geq 5\%$ for roofs without transversal end laps.
 - $\geq 7\%$ for roofs with transversal end laps (joining two panels longitudinally).
- **Typical Load Capacity:** Available in thicknesses of 50, 80, 100, 120, and 150 mm. For a 100 mm thick panel spanning 2.00 m between structural purlins, the panel supports an allowable service load of up to 1.65 kN/m^2 (combined wind pressure/suction and snow load), limiting mid-span deflection to $L/200$.

2.3 Acoustic / Sound-Absorbing Panel: “RockAcoustic”

A cutting-edge solution designed specifically for industrial noise control, interior acoustic conditioning, and highway or industrial sound barriers.

- **Inner Facing Design:** Perforated steel inner liner with a symmetric R3T5 pattern (3 mm diameter round holes spaced at 5 mm intervals), yielding approximately a 33.5% open perforated area. A black, hydrophobic, non-woven glass tissue layer is placed between the rock wool core and the perforated steel sheet to prevent mineral fiber migration into the environment.
- **Industrial Applications:** Engine rooms, compressor enclosures, co-generation plants, acoustic partitions in noisy manufacturing facilities, and duct silencers.
- **Acoustic Performance:**
 - **Sound Reduction Index (R_w):** Ranges from 32 dB (50 mm thickness) to 35 dB (120 mm thickness) certified under *EN ISO 140-3*.
 - **Sound Absorption Coefficient (α_w):** Achieves up to $\alpha_w = 0.90$ (Class A absorber according to *EN ISO 11654*), effectively absorbing mid-to-high frequencies typical of industrial machinery.

3 Fire Performance Tables

The fire resistance of our rock wool sandwich panels is tested and certified under the stringent European standard **EN 13501-2**, assessing **E** (Integrity) and **I** (Insulation) criteria.

The table below outlines the fire resistance rating as a function of panel thickness for self-supporting wall applications (vertical alignment) and roofs (horizontal alignment), using a rock wool core with a nominal density of $\rho = 100 \text{ kg/m}^3$:

Table 1: Vertex Rock Systems Fire Resistance Ratings

Thickness (mm)	Resistance Class (EN 13501-2)	Time Limit (Integrity & Ins.)	Max Testing Load Status	
50	EI 60	≥ 60 minutes	Self-supporting loadbearing)	(Non-
80	EI 90	≥ 90 minutes	Self-supporting loadbearing)	(Non-
100	EI 120	≥ 120 minutes	Self-supporting loadbearing)	(Non-
120	EI 120	≥ 120 minutes	Load-bearing (Under reduced load)	
150	EI 180	≥ 180 minutes	Self-supporting loadbearing)	(Non-
200	EI 240	≥ 240 minutes	Self-supporting loadbearing)	(Non-

Note: All specified ratings strictly require the assembly to be executed in compliance with the manufacturer's guidelines regarding fastener type, size, layout density per support, and joint sealing with approved intumescent silicones.

4 Basic Installation and Handling Guide

To ensure that the structural, thermal, acoustic, and fire resistance properties remain intact upon completion of construction, installers must strictly adhere to the following key guidelines:

4.1 Mechanical Fasteners and Load Distribution

- **Dimensioning:** Fasteners must be sized to withstand the calculated local wind suction loads in accordance with Eurocode 1 (*EN 1991-1-4*).
- **Essential Hardware:** Use stainless steel or carbon steel fasteners with a minimum C4 corrosion protection coating. Each external fastener must incorporate a weatherproofing saddle (rider) or an EPDM elastomer washer (minimum diameter of 16 mm for walls and 19 mm for roofs).

4.2 Joint Sealing and Weatherproofing

- **Air and Vapor Tightness:** The tongue-and-groove joint of longitudinal panel connections must be reinforced by applying continuous beads of high-quality polyurethane or butyl mastic sealant inside the joint recess prior to locking the panels together.
- **Fire Barrier Sealing:** For fire-rated compartments requiring classifications higher than **EI 120**, internal joints and junctions with the main structure must be sealed using approved intumescent acrylic or silicone sealants designed to expand at temperatures above 180°C.

4.3 Modulation, Tolerances, and Assembly

- **Substructure Planarity:** Prior to panel installation, verify that the alignment and flat tolerance of the supporting steel or concrete structure does not exceed ± 2 mm over a 2.00 m span.
- **On-Site Cutting:** Use high-quality jigsaw or circular saws with low-speed settings and blades specifically designed for soft metals. The use of angle grinders (abrasive discs) is strictly prohibited as the generated heat destroys the zinc galvanization layer and the paint coating.

4.4 Safe Storage on Site

- **Stacking and Slope:** Store panel bundles on a level surface using supporting timber blocks spaced at a maximum of 1.50 m. Ensure a minimum slope of 2% in the stack setup to prevent ponding water.
- **Protective Film Removal:** The protective plastic film must be peeled off within 45 days of delivery, or within 7 days if panels are directly exposed to solar radiation on site.